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# Hanlin Tian
Postgraduate Researcher

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## Education

**MPhil. in Civil and Environmental Engineering**, *Transport System and Logistics Lab, Imperial College London* **Expected 2024**
Research Focus: Autonomous Vehicles, Computer Vision, Large Language Models, Reinforcement Learning
**Master of Science in Computer Engineering**, *New York University* **October 2019 - June 2021**
Relevant Coursework: Advanced Machine Learning, Algorithmic Machine Learning and Data Science, Information Theory
**Bachelor of Engineering in Computer Science**, *Shandong University* **September 2015 - June 2019**
Relevant Coursework: Advanced Programming Language, Pattern Recognition
**Undergrad Summer Visitor**, *Stanford University* **June 2017 - August 2017**

## Research Experience

**Bézier Everywhere All at Once: Learning Drivable Lanes as Bézier Graphs** **October 2023**
- *Hugh Blayney, **Hanlin Tian**, Hamish Scott, Nils Goldbeck, Chess Stetson, Panagiotis Angeloudis*
- Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2024, pp. 15365-15374
- Developed a method for large-scale extraction of lane topology from aerial imagery, utilizing Bézier curve parameterization and a transformer-based model.
- Achieved competitive results on the Urban Lane Graph dataset, producing realistic lane graphs with minimal input, and shared the code for public use.

**Large (Vision) Language Models for Autonomous Vehicles: Current Trends and Future Directions** **July 2024**
- ***Hanlin Tian**, Kethan Reddy, Yuxiang Feng, Mohammed Quddus, Yiannis Demiris, Panagiotis Angeloudis*
- Submitted to **IEEE Transactions on Intelligent Transportation Systems**
- Provided a comprehensive review of L(V)LM integration in autonomous vehicles, assessing methodologies, outcomes, strengths, and limitations of current research.
- Proposed future research directions focused on L(V)LM integration with AV systems, addressing regulatory and ethical challenges, and exploring potential in V2X communication.

**Multimodal Learning for Traffic Risk Prediction: Combining Aerial Imagery with Contextual Data** **October 2024**
- ***Hanlin Tian**, Yuxiang Feng, Mohammed Quddus, Yiannis Demiris, Panagiotis Angeloudis*
- Submitted to **IEEE Open Journal of Intelligent Transportation Systems**
- Introduced a multimodal deep learning framework that integrates aerial imagery, building footprint data, and traffic flow information to enhance traffic risk prediction at urban intersections.
- Achieved an Intersection over Union (IoU) score of 0.4052 and a Root Mean Square Error (RMSE) of 0.0907 with the DeepLabV3+ model incorporating both building and traffic data.

**Enhancing Autonomous Vehicle Training with Language Model Integration and Critical Scenario Generation** **June 2024**
- ***Hanlin Tian**, Kethan Reddy, Yuxiang Feng, Mohammed Quddus, Yiannis Demiris, Panagiotis Angeloudis*
- arxiv.org/abs/2404.08570
- Introduced CRITICAL, a novel framework for AV training and testing, generating scenarios to address learning gaps in RL agents.
- Demonstrated enhanced performance and safety in AV systems, as evidenced in PPO and HighwayEnv simulations.

**Impact of Building-Induced Visibility Restrictions on Intersection Accidents** **May 2023**
- ***Hanlin Tian**, Yuxiang Feng, Wei Zhou, Anupriya, Mohammed Quddus, Yiannis Demiris, Panagiotis Angeloudis*
- Presented at the **TRB Annual Meeting**, TRBAM-24-02409.
- Developed a framework for evaluating urban junction safety using OSM and remote sensing data.
- Utilized a Graph Neural Network for comprehensive risk assessment, with real-world case studies validating the methodology.

**Vehicle-to-Vehicle Collaborative Navigation for Autonomous Vehicles in Occluded Environments** **October 2022**
- *Leandro Parada, **Hanlin Tian**, Jose Escribano, Panagiotis Angeloudis*
- Presented in **ITSC** 2023, DOI: 10.1109/ITSC57777.2023.10422217.
- Developed a V2V collaborative control methodology using Proximal Policy Optimisation for enhanced safety in occluded environments.
- Validated the approach in a CARLA-based simulation, showing improved performance over traditional strategies.

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**Safety Standards for Autonomous Vehicles: Challenges and Way Forward** — **May 2023**

- *Elias Nassif,* ***Hanlin Tian****, Eduardo Candela, Yuxiang Feng, Panagiotis Angeloudis, Washington Y Ochieng*
- Presented at **ITSC** 2023, DOI: 10.1109/ITSC57777.2023.10422153.
- Analyzed existing safety standards and proposed a balanced approach for AV standardization.
- Developed principles and a strategic plan for promoting standards adoption in the AV industry.

**The Robust Semantic Segmentation UNCV2023 Challenge** — **October 2023**

- ***Hanlin Tian****, Kenta Matsui, Tianhao Wang, Fahmy Adan*
- Presented in **ICCV** Workshops 2023, article on pp. 4618-4628.
- Secured 6th place in the UNCV2023 Challenge.
- Implemented a snow simulation data augmentation technique for improved segmentation in winter conditions.
- Integrated the Xception architecture into our segmentation model for enhanced performance.

**Active Crowd Analysis for Pandemic Risk Mitigation for Visually Impaired Individuals** — **July 2020**

- *Samridha Shrestha, Daohan Lu,* ***Hanlin Tian****, Qiming Cao, Julie Liu, John-Ross Rizzo, William H. Seiple, Maurizio Porfiri, Yi Fang*
- Presented at **ECCV** Workshops 2020, article on pp. 422-439.
- Developed a smartphone-based neural network for crowd density detection, aiding visually impaired individuals with social distancing.
- Utilized real-time crowd analysis for effective pandemic risk mitigation.

**Meta-DCP: Meta-Learning and Deep Learning for 3D Point Cloud Registration** — **May 2021**

- Master's Thesis research.
- Employed Meta-Learning and Deep-Learning techniques for 3D point cloud registration.
- Demonstrated lower loss on unseen point cloud categories compared to traditional methods.

## Activities

**Research Assistant, Imperial College London** — **September 2023 -**

- Engaged in the DeepSafe initiative, focusing on enhancing synthetic data for training AVs in edge-case scenarios to improve performance and reliability.

**Research Intern, dRisk** — **June 2023 - September 2023**

- Developed advanced lane graph generation techniques using transformer architectures and Bezier curves for enhanced accuracy and system performance.

**Research Assistant, Zhejiang University** — **October 2022 - January 2023**

- Worked with Professor Simon Hu at the Transport Systems & Environment Lab on assembling and evaluating autonomous vehicle models, focusing on hardware assembly and performance analysis.

**Research Assistant, New York University** — **August 2020 - July 2021**

- Contributed to the publication "Active Crowd Analysis for Pandemic Risk Mitigation for Visually Impaired Persons" and the development of the "Virtual Touch" assistive platform for visually impaired individuals.
- Authored a Master's thesis, "Meta-DCP," with potential for conference submission.

**Software Development Engineer, Chinese Academy of Science** — **February 2019 - July 2019**

- Developed a comprehensive Industrial Vision System and designed a neural network for industrial part inspection using TensorFlow.

**Summer Research Assistant, Washington University in St. Louis** — **July 2018 - September 2018**

- Created an Autoencoder for image compression and denoising, achieving promising results in both applications.

## Expertise and Skills

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|---|---|
| **Tools and Languages** | Python, C++, Git, LaTeX, PyTorch, TensorFlow, R, MongoDB |
| **Research Experiences** | Machine Learning, Computer Vision, Meta-Learning, Reinforcement Learning |
| **Communication** | English, Cantonese, Chinese |